

2.1 - THE PRIZE

Amsterdam, 2050.

My hands rest on the iron railing of a bridge over the Keizersgracht – the Emperor's Canal – named after Maximilian I, who dug it in 1615. But its purpose was not to serve the emperor but to serve merchants who insisted on having their packages delivered by boat to their front door. It tells you something about the power dynamics in the Netherlands. Four centuries later the façades still lean forward, so the furniture or merchandise being hoisted up on beams sticking out from the top of the façade wouldn't scrape the brickwork. The Dutch have been going against convention if that solves practical problems for a long time. It's late afternoon and light catches the tall windows of one particularly picturesque house across the water and turns them briefly, outrageously, gold.

I ache in the places that I used to play, as [Leonard Cohen](#) once put it, and at eighty-five that covers a lot of territory. But I'm happy. I'm happy watching a tour boat glide through water so clear I can see fish darting along the canal bed. I can even check that there are no drowned bicycles down there — which, for Amsterdam, is quite astonishing.

Two teenage girls are giggling hysterically at something they're sharing through their VR glasses. A couple of Americans at the terrace below me along the water are delivering a full-volume verdict on the *bitterballen*. "It's like a croquette but *angry*," one of them announces, and I smile, because that's exactly right, and because some things apparently never change no matter how much technology you throw at them.

I breathe in deeply. I let the clean air go all the way down in my lungs while I'm trying to get a scent for what's happening down there in the canal. Hanging there over the railing of this bridge I'll probably look like a golden retriever hanging out of a car window. But decorum left me somewhere around my sixtieth birthday. I don't miss it much.

And it is so quiet. I can hear the water against the canal wall. I can hear the Americans. I can hear blackbirds in the lindens, singing as if the entire city were built as an amphitheatre for their personal recital. The tour boat passes close now, making a soft hum that could be a large cat with a clear conscience.

Small vehicles weave past me on the bridge — three-wheeled, roofed, barely wider than a bicycle. They thread between pedestrians at walking speed. One swerves patiently around a child who has stopped dead to stare at a dog with an expression suggesting she has just discovered dogs exist. The vehicles cope without problems. They don't honk. They silently flow around obstacles with the grace of water going around a rock.

I watch them for a while, this silent choreography in a clean city, and I feel something in my chest that might be joy, satisfaction, or even pride. Pride of a stubborn old man who spent decades arguing that this was possible and then — against what felt for a long time like the combined resistance of every energy company, grid operator, government official and tenured professor on the continent — lived long enough to stand here leaning on this bridge and watch it happen.

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Yesterday, or was it last week? Things are a little bit foggy these days. I had a virtual reality call with my friend Amara, my contact on the ground in Kenya. The VR is so good now that I could almost smell the red earth around her home. She walked me around her two hectares while small robots scurried between the rows of maize — squat, rugged machines about the size of a Jack Russell, weeding with surgical precision. She talked to them the way you'd talk to a slightly dim but willing dog. "Kidogo, you missed one." Kidogo reversed and got the weed.

One of the robots was off fetching veterinary supplies from the village depot, twenty kilometres away. It would be back within the hour, Amara said, and shrugged as if dispatching an autonomous robot on a forty-kilometre round trip were as unremarkable as sending your kid to the corner shop.

Her roof had solar panels — maybe eight square metres. More than enough, she said, for the fridge, the induction stove in the kitchen, LED light in every room... the amenities of modern civilization in the middle of nowhere. Her kids were doing homework at the table with the focused irritation of teenagers everywhere who would rather be anywhere else. The house was cool and clean and quiet, and Amara was showing me her new grain dryer, which ran on a heat pump, and which she described as the best thing after robots.

And as I remembered Amara I thought of my parents. I was now older than they had ever been. They had missed this new reality — this Amsterdam and this Amara — by decades. But they would have been so happy. My father, who taught me that it was the job of a man to avoid suffering for those less able to stand up for themselves. My mother, always so gentle but fierce when it came to people in need, because she knew what not having enough looked like. They both knew that you measure any system by how it treats the people at the bottom. Not the average. The bottom.

My father would have asked Amara questions and silently beamed with joy at her improved standard of living. My mother would have cried a little, and clung to me, and told me: this was how it should be.

And thinking of them, my thoughts turned to the conference I had attended only yesterday. A conference at – what was it again – ah yes Lappeenranta. The city where [LUT University](#) sits on the shore of Lake Saimaa — the fourth largest lake in Europe with birch forests in every direction. I had been at the annual energy transition conference. Halls full of young PhD students figuring out what to do with their lives, milling around poster sessions, drinking terrible conference coffee, looking faintly terrified about their upcoming presentations. Some things never change.

Tucked into the Wednesday afternoon programme was a session that [Michael Liebreich](#) had christened "Old Geezers Doing a Victory Lap." Our little reunion.

For over four decades, we had insisting, with varying degrees of diplomacy, data, and recklessness, that the energy transition would go faster and further than the mainstream believed, and especially faster and further than the mainstream *experts* believed. The mainstream had spent the last four decades catching up with us at the speed bureaucracy

discovering email. But we had been there first, and this was our victory lap, and God help anyone who tried to take it from us.

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The session opened with climate. [Glen Peters](#) of CICERO, Zeke Hausfather of Berkeley Earth, and [Justin Ritchie](#), who back in the 2020s had done the meticulous, thankless, occasionally career-limiting work of demonstrating that the most extreme climate scenarios were built on an [implausible assumption of a global return to coal](#). For years, the worst-case scenario — RCP8.5, which projected a six-fold increase in global coal consumption per capita by 2100 — had been treated as "business as usual" by most researchers, hundreds of newspaper headlines, and approximately every climate anxiety article ever posted on social media. Justin first, Glen and Zeke later, had pointed out that this was simply wrong. Michael and I had been piling on. Michael even invented a hashtag for it: [#RCP85isBollocks](#). A lot of people told me that is how they learned what *bollocks* meant. And it was bollocks, because coal was already declining when the scenario was written. The whole edifice of the worst climate horror scenarios was built on a six-fold expansion of something everybody could already see was shrinking.

They and we had not been popular for saying this out loud. Climate sceptics had seized on the work to argue that climate change wasn't a problem. Climate activists had accused us of downplaying the threat. We had, in fact, been doing neither. We had been doing science, which is the radical act of checking whether your assumptions match the data and reporting the answer even when it is not what you would like it to be and even if both sides of the political divide would rather you hadn't asked.

Their presentation in Lappeenranta was a shared exhale. Emissions had peaked in the late 2020s. The trajectory now pointed to about 1.3 degrees of warming by 2100. Not zero damage — that ship had sailed long ago — but a planet very different from the catastrophe that once seemed inevitable. Glen, who had spent his career parsing emissions data with the affect of a man auditing a particularly hostile spreadsheet, allowed himself a smile. It was like watching a glacier do something unexpected.

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After them came [Jenny Chase](#) of Bloomberg New Energy Finance (BNEF), who had been tracking solar since the industry's entire global capacity could have fitted on a large car park. Jenny's talk was simple. When we started, solar was an ambition; now it had completely taken over the world. Not as a supplement. Not as one option among many. As *the* dominant energy source on the planet. I had discovered that almost half a century ago — around 2006 but it was still surprise to some people. She showed a chart of cumulative installed capacity that started as a timid pencil line in the early 2000s and ended as something you'd need to stand well back from to see the top of. I caught her eye and mouthed *told you so*. She rolled her eyes with the precision of a woman who has been rolling her eyes at overconfident men in the energy sector for thirty years and has got rather good at it. Her analysts were the best in the business but she not-so-secretly agreed with me that they often "chickened out" (her words) when predicting rapid adoption.

Then our host – and my friend of five decades – [Christian Breyer](#) took the stage. Christian had spent his entire career at LUT modelling [100% renewable energy systems](#), even when the phrase alone was enough to get you disinvited from serious policy discussions. He had published study after study showing it was feasible, viable, and necessary. He had been ignored, disputed, patronised, reluctantly cited, and finally vindicated — the full academic stations of the cross, in the correct order. Now retired but showing no signs of stopping — the sort of man who will give a presentation about solar energy at his own funeral, and will still make the audience stay for the Q&A — Christian explained with undimmed enthusiasm that solar was *still* finding new niches. Even wind power, he noted, smiling in a way that made the wind energy people in the audience shift in their seats, was now losing market share to solar.

And then his main focus: direct air carbon capture, powered by surplus solar, pulling CO₂ back out of the atmosphere. Slowly. Less expensively by the year. More steadily and forcefully by the year. That was what had made 1.3 degrees by 2100 — and falling — possible. Christian said this as if reporting a minor correction to a spreadsheet.

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Then [Michael Liebreich](#). Now eighty-seven. No less opinionated in his talk than at thirty-seven, which is saying something if you know Michael. Here was the man who had founded New Energy Finance before it became Bloomberg New Energy Finance, and before renewable energy had been treated seriously. The man who had created the [hydrogen ladder](#) — the simple graphic that had done more to deflate the hydrogen hype than a thousand white papers. He was doing what he had been doing for four decades: explaining that he had been right about everything all along, and doing it so entertainingly that you didn't mind. Or at least that you didn't interrupt.

The hydrogen ladder had been heresy when he first published it around 2020. Hydrogen enthusiasts — and there had been many, most of them holding freshly printed government strategy documents — had wanted hydrogen everywhere. Heating homes. Propelling cars and and busses. Anything that would justify the massive infrastructure investments their employers were lobbying for. Michael had pointed out that for most of these applications, you were better off just using electricity directly. Converting electricity to hydrogen and then back again was a costly way to throw away half of the electricity. Hydrogen made sense for seasonal storage, for some industrial feedstocks, for a few niche applications. For everything else it was, as Michael had loved to say, a [hydrogen soufflé](#): a lot of hot air, destined to collapse the moment you looked at it properly.

Of course he had been right. Actually many people in this room had been right, although they hadn't produced this easy-to-understand ladder. By 2050, hydrogen occupied exactly the niches most of us centenarians in the reunion had predicted, and approximately none of the ones its loudest champions had promised. Michael covered this in about five minutes, which for him was an act of restraint bordering on the heroic.

Then he turned to what everyone in 2025 had called the "hard-to-abate" sectors — and which he had [written about years ago on his Substack](#) under the heading that these sectors were not, on closer inspection, all that hard to abate. Cement, which once accounted for

roughly 8% of global CO₂ emissions. Steel. The poster children of the "some things just can't be electrified" argument — an argument that had aged about as well as "heavier-than-air flight is impossible." Now we had green hydrogen for the direct reduction of iron ore and electric arc furnaces, running on solar electricity, had replaced their coal-fired ancestors at lower cost. Not overnight — these were slow-moving industries run by cautious people who considered a thirty-year planning horizon adventurous — but decisively. Michael dispatched all of this with the brisk efficiency of a man who had been saying "I told you so" professionally for fifty years and had optimised the process.

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The session turned to transport. [Colin McKerracher](#) of BNEF was now taking the stage. He was the youngest person in our session, which in this company meant about seventy-five. We called him "the teenager." Like me, he had been tracking electric vehicles since before most car companies took them seriously. I should note, and I will note, because nobody else is going to and this is my book, that my friend [Maarten Steinbuch](#) and I had been making the case for electric cars and even electric trucks since 2009. Five years before Colin started working on them, I had already written a (Dutch) book on electric cars. Maarten had sat in the frunk of his Tesla and was telling anyone who'd listen – and the others too – that his great-great-great-grandfather had helped found the automotive school at [TU Eindhoven](#) for combustion vehicles, and that he, Maarten, has created an automotive master striving to make combustion obsolete. We were early. We were weird. We were right.

Colin's presentation was almost anticlimactic: electric cars had fully displaced combustion in new sales. Trucks too. These were not predictions. They were history. What interested Colin now — the reason he'd dragged himself to Finland, although we kept pointing out that he'd only recently retired and should really stop complaining about his joints — were the last holdouts: long-haul shipping and long-haul aviation.

Shipping had gone through two acts. First, batteries conquered everything under about 1,500 kilometres. Weight doesn't matter much when your vehicle already displaces ten thousand tonnes of water, and both iron-air and sodium batteries had become obscenely cheap. For longer voyages, ships had at first switched from diesel to ammonia and methanol synthesised from renewable electricity. But then — and in hindsight this was obvious, which is the most annoying kind of obvious — enterprising women and men started building offshore wind farms that functioned as a kind of car charger on sea, but larger. Same principle as the solar carport in your driveway with its car charger, now scaled up by a factor of about a million. Once you could top up mid-ocean, batteries started winning the long routes too. The ammonia and methanol producers were just as unhappy as the diesel producers had been fifty years earlier. And everybody at the conference felt about as sorry for them as they had felt for the diesel producers in that bygone era.

Maarten took the stage after Colin. Short-haul flights had gone battery-electric in the 2030s. Longer distances had initially switched to synthetic kerosene — electro-fuels. But battery energy density kept climbing, and since electric motors are roughly three times more efficient than jet engines, there came a point where even batteries that weighed twice as much as kerosene were winning on energy delivered to the propeller. Even long-haul was going electric. Maarten explained this with the satisfaction of a man who had been

fantasising about this for most of his career and had now been able to board his once fantasy at Schiphol on his flight to the conference.

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After lunch the mood shifted as [George Monbiot](#) took the stage. Eighty-seven years old, still writing, still angry about large corporations dominating the little man, still angry about animal abuse, and most of all still angry about people demolishing wildlife in England. Still capable of making a room full of energy specialists feel like they'd been staring at the problem through a keyhole. He'd spent the morning inspecting the birch forest around the university and had returned in high spirits, which for George means newly furious.

He shamed us. That is the only word for what he did — and most of the room was already on his side, which somehow made it worse. He reminded us that in 2025 humanity had been running what amounted to the largest system of deliberate torture ever devised, and had euphemistically been calling it *factory farming*. He cited [a study from Heilongjiang Bayi Agricultural University](#) where researchers had tried to find the most effective way to induce clinical depression in pigs — for medical research, not cruelty, although George left a pause there that did a lot of work. The answer? Confine the pig so tightly it can't turn around, and isolate it from other pigs. In other words: standard factory farming practice. The exact conditions in which billions of animals — creatures more intelligent and more social than the dogs we'd go to prison for treating the same way — had spent their entire lives. We had known this. We had known it and we had eaten our lunch and we had not thought about it very hard.

Then George pivoted from animals to land. The staggering share of the Earth's surface that had been devoted to growing feed for the livestock we tortured and consumed. The forests razed. The biodiversity annihilated. He'd been writing about this on and off since [Regenesi](#)s came out in 2022, and by 2050 the data was vindicating him with an almost embarrassing thoroughness. But George wasn't here to take a bow. George was here to make sure we remembered how bad it had been, because George does not believe in letting people feel good about themselves when there is still work to be done, and there is always still work to be done.

[Jonathan Foley](#) went next and was a shade more analytical, which after George was like stepping from a cold shower into a heated towel. Jonathan had run [Project Drawdown](#) for years — the most comprehensive effort to rank and quantify climate solutions. He piled on the data about how terrible the food system had been, but then he turned the corner: reduced food waste, plant-based diets, precision agriculture, cultivated meat — together, they had freed up vast amounts of land. And that land was rewilding. Forests regrowing. Grasslands recovering. Species coming back from edges they should never have been pushed to. Jonathan showed maps of rewilded territory spreading across the continents and I found myself blinking hard, which at eighty-five I will blame on dry air and not on anything as undignified as the urge to cry.

[Elliot Swartz](#) from the [Good Food Institute](#) presented on cultivated meat — real animal protein grown from cells, no animal required. By 2050, cultivated and plant-based proteins had taken most of the global market. Only about ten percent of meat was still from animals.

Eating it was becoming, as I like to put it, like wearing fur. Not illegal. Just not something you brought up at dinner parties.

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The last speaker before me was Kingsmill Bond of the [Rocky Mountain Institute](#), who had spent decades mapping what happens when the fossil fuel economy unravels. His concept of the electrostate — the geopolitical order that emerges when energy is domestic sunshine instead of imported oil — had gone from a provocation to a fact of life. Nations that had once been petrostates now had the geopolitical leverage of a country whose chief export has gone out of fashion. You cannot embargo the sun. You cannot put sanctions on the wind. Energy politics had shifted from extraction to manufacturing, from scarcity to abundance, from a few nations sitting on fossil reserves to every nation sitting under the same sky.

And then it was my turn.

Of course, I said, looking out at the audience — half PhD students who were wondering why these old people kept congratulating each other, half government officials who were getting slightly annoyed about exactly that — of course everybody in this room thinks holons are obvious. Of course the grid is organised as nested systems, because it is like any other complex adaptive system. Of course every home is an energy node. Of course your neighbourhood balances its own supply and demand before bothering the wider grid. Of course, of course, of course.

But it's good to take a moment, I said, to realise how much has changed in the last decades. Because forty years ago this was still a very strange idea. The grid was centralised, and power flowed one way — from big plants to dumb consumers. The notion that your home, your street, your city could each function as a semi-autonomous energy cell was loved but uncomprehended by social scientists, and struck most grid engineers as naive, most economists as irrelevant, and most policymakers as pie-in-the-sky thinking.

I'd been one of the first to insist that this was the missing piece. The organisational architecture without which solar panels and batteries and smart inverters would remain clever gadgets rather than the foundations of a new energy civilisation. And I'd been one of the early voices — not the only one, but one of the early ones — shouting into the void that experts were massively underestimating the growth of solar, electric cars, electric trucks, vehicle-to-grid, stationary batteries. I had the receipts.

For most people in that room, the changes to flying, shipping and food were much more tangible, and seemed much more important, than anything I was about to say. And as I was standing there I secretly thought that my most important contribution might be none of those things. It might be the quiet, stubborn, slightly obsessive insistence that the energy system should be structured the way nature structures everything: as nested, semi-autonomous cells. Each one whole in itself and part of a larger whole. Holons. Borrowed from Arthur Koestler, applied to the electricity grid, dismissed for years, vindicated by events.

That was arrogant. I knew it was arrogant. I thought it anyway. You get to be arrogant at eighty-five. It's one of the few perks.

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I blink.

The fish are gone. The water is opaque, the colour of over-stewed tea, and I'm sure there is many bicycles down there in the canal. The houses are leaning forward and still beautiful but coated in a familiar thin film of grime. The air is not clean but has the faint, metallic, taste of exhaust. The signature of a civilisation still running on controlled explosions. I know breathing this air will statistically shave two years of my life and probably killed my mother and [millions of other people](#) far more quickly than that. But that's just how it is.

And the noise. God, the noise. A scooter tears past with the whine of a two-stroke engine designed by someone who hates ears. A delivery van shakes the bridge. The tour boat churns through the canal trailing diesel behind it like a bad reputation. The Americans might still be at the terrace but I can't make out their argument any more. The blackbird might still be singing, but I'll never know.

It was a daydream. All of it. The conference, the electric plane, Amara and her robots, the clear canal, the clean air, the quiet.

It is 2026 again. My parents have passed away, but the future they told me to build — one measured not by the comfort of the comfortable, but by what it does for people like Amara — is not there yet.

Right now, today, [770 million people](#) have no access to electricity. Over [two billion people](#) still cook over wood, charcoal, dung, or coal — fires that fill their homes with smoke that kills roughly [3.2 million people a year](#), mostly women and children, and we have known this for decades. Children study by kerosene lamps that [cost more per lumen than solar-powered LEDs already cost today](#), and those fumes are among the most concentrated carcinogens you can expose a child to.

The air outside is not much kinder. Another 4.2 million people a year die from outdoor air pollution. [Ninety-nine percent of the world's population](#) breathes air dirtier than the WHO says is safe. Not people in particularly polluted cities. Nearly everyone, everywhere. In rich countries we still burn fossil gas in our kitchens — an arrangement that [kills nineteen thousand Americans a year](#) and around [forty thousand Europeans](#), though most of them will never know what did it. India loses 1.7 million people a year to air pollution with fossil fuels responsible for more than 750,000 of those. Meanwhile we keep building gas pipelines that we will have to close early, and nuclear plants that take a decade longer to build than the renewables that would have done the job — a decade during which the existing fossil plants keep running, and keep killing people, which is a line item nobody seems to want to put on nuclear's ledger.

Land tells the same story. Roughly half of the planet's habitable land is given over to farming, and [around 80% of that farmland feeds livestock](#) rather than people. Extinction rates are running [ten to a hundred times the natural background](#). The Amazon is still being cleared for soy to feed pigs in Europe and cows in China. Pigs we still torture using “factory farming”. Petrostate dictators are still funding wars while the world looks away, because we

still need what they have. The grid is still centralised, fragile, increasingly expensive and overdue for reinvention.

I'm standing on this bridge and I'm furious. This is not happening on my shift.

Getting from here to that daydream — from murky water to clear, from kerosene to solar, from a grid designed for the fossil fuelled century we left behind to one fit for to support the energy transition in this one — is what the rest of this book is about.